

Efficient implementation of image representation, visual geometry group with 19 layers and residual network with 152 layers for intrusion detection from UNSW-NB15 dataset

Youssef F. Sallam¹  | Samy Abd El-Nabi² | Walid El-Shafai^{1,3}  |
Hossam El-din H. Ahmed¹ | Adel Saleeb¹ | Nirmeen A. El-Bahnasawy⁴  |
Fathi E. Abd El-Samie^{1,5}

¹Department of Electronics and Electrical Communications, Faculty of Electronic Engineering, Menoufia University, Menouf, Egypt

²Alexandria Higher Institute of Engineering & Technology (AIET), Alexandria, Egypt

³Security Engineering Lab, Computer Science Department, Prince Sultan University, Riyadh, Saudi Arabia

⁴Department of Computer Science and Engineering, Faculty of Electronic Engineering, Menoufia University, Menouf, Egypt

⁵Department of Information Technology, College of Computer and Information Sciences, Princess Nourah Bint Abdulrahman University, Riyadh, Saudi Arabia

Correspondence

Youssef F. Sallam, Department of Electronics and Electrical Communications, Faculty of Electronic Engineering, Menoufia University, Menouf, Egypt.
Email: yousef1616@gmail.com

Abstract

The Internet offers humanity many distinctive and indispensable services, whether for individuals or for institutions and companies. This great role has attracted the Internet attackers to develop their mechanisms to capture and obtain the data by illegal methods. This growth in the number of cyber-attacks made scientists in a real challenge, to find advanced methods to face this danger. Due to the shortcomings of traditional data security means such as firewalls, encryption, and so forth, the motivation became to develop alternative systems to detect smart attacks. Intrusion detection systems (IDSs) have made remarkable progress in cyber-security. They monitor the traffic in real time and continuously to detect the network attacks, giving alerts to the network administrator. In this article, two IDSs are introduced based on principles of transfer learning (TL) with convolutional neural networks. Our systems are built using the visual geometry group (VGG19) and residual network with 152 layers (ResNet152). UNSW-NB15 intrusion detection dataset is used to evaluate the models. The proposals achieve high levels of precision, recall, and F1_score as 99%, 99%, and 99%, respectively. These achievements prove the efficiency of the proposed models in capturing cyber-attacks with low alert rates.

KEYWORDS

CNN, IDS, ResNet152, transfer learning (TL), UNSW-NB15, VGG19

1 | INTRODUCTION

The diversity of data allows it to meet the needs of people of all kinds and objectives. Data confidentiality is extremely important to both individuals and institutions alike. Data security has become one of the most important things in the lives of individuals and institutions.

From this point of view, scientists are always working on developing advanced systems to preserve data from loss, whether it was by theft, destruction, unlawful disclosure or any other unethical conduct. The usual data security systems such as anti-viruses, encryption and firewalls, have become unable to cope with the huge development in the size and

quality of cyber-attacks. This shortcoming in the work of these systems prompted cyber-security scientists to intensify their efforts towards finding additional and effective solutions to detect these attacks.

Intrusion detection systems (IDSs) are one of the best benefits of researchers' works as they put a constant eye on the behavior of people and data in the network, to detect any unusual behavior that might demonstrate that an assault will occur, is occurring, or has occurred. An IDS is made out of equipment and programming components.¹ Cyber-attacks vary in many forms, including phishing, DoS, malware, probing, man in the middle (MitM), R2L, U2R, and many other types.² This diversity of attacks is offset by another diversity in the types and methods of detection.

The IDSs are classified based on many factors, the most important of which are detection mechanism and scope of work. As for the detection mechanisms, they are divided into three types:

Misuse or signature-based detection: In this type, any behavior similar to what is registered in the signature database of previous attacks is defined as an attack. One of its most important features is the accuracy of detecting attacks similar to the attacks that were previously detected, but it has shortcomings in detecting the types of modern attacks.³

Anomaly-based detection: The idea of this type is based on modeling of the natural behavior of data traffic and considering any contrary behavior as a malicious behavior. Its most important advantage lies in the ability to identify new attacks, but it suffers from an increase in false alert rates.⁴

Hybrid detection: In this type, a mixture of the two previous types is used in order to exploit the advantage of one of them in detecting previous attacks and the other in identifying new ones.

Referring to the division of IDSs based on their field of work, they come in three types. The first is the host-based IDS (HIDS), in which the system detects the abnormal behavior in relation to user activities on a single device.⁵ The second type monitors the activities of users at the level of the network as a whole, and that is why it is called the network-based IDS (NIDS).⁶ Finally, in the hybrid IDS, monitoring is made for user activities on their devices separately once, and at the level of the network as a whole once again. These types are shown in Figure 1.

Since it was started by J. Anderson in the 1980s, many proposed models for the work of IDS have been presented.⁷ This diversity is intended to reach intelligent systems with high accuracy of data classification, while reducing the percentage of false alerts, without increasing the cost and complexity of the systems. Those in charge of these systems used several methods. The most famous of them are machine learning (ML) systems such as decision tree (DT), multi-layer perceptron (MLP), logistic regression (LR), Naive Bayes, k-nearest neighbors (kNN), SVM, random forest (RF), AdaBoost and so forth.⁸ These systems have offered distinct accuracy levels for a period of time, but with the massive increase in the volume of data, they have faced many difficulties. With the advent of deep learning (DL) methodology, there has been a huge leap in the fields of data classification, and IDSs have benefited from this boom. The DL systems addressed problems related to the increase in the volume of data, but in contrary, their efficiency increased significantly as the volume of data increased.⁹

Convolutional neural networks (CNNs) represent one of the most famous types of DL due to their ability to auto-extract the best features from data, and classify them without the need for a separate feature extraction stage.¹⁰ This advantage increased the efficiency of the IDS systems on the one hand, but increased complexity on the other. So, CNNs are the main foundations of our proposed systems.

The remainder of this article is distributed as follows. Some related works are summarized in Section 2. In Section 3, our proposed models are explained. Simulation results are given in Section 4. Finally, the summary and conclusion of our research and future work are given in Section 5.

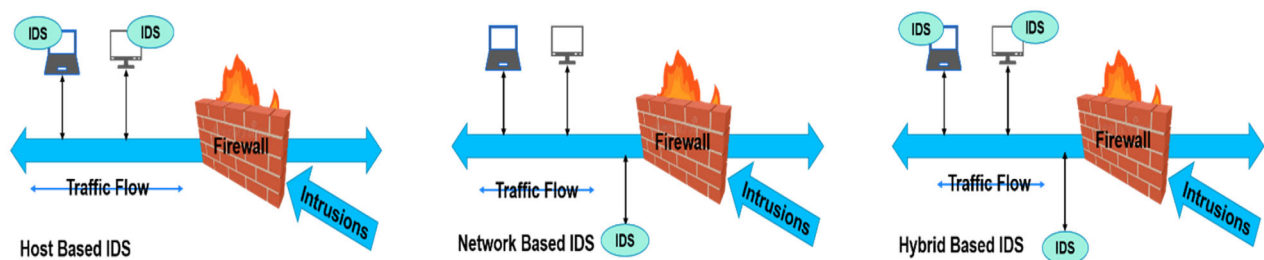


FIGURE 1 IDS based on working location

2 | RELATED WORK

The diversity of cyber-attacks and the exponential increase in their number have made it a constant challenge to produce intelligent systems to detect these attacks. One of the most important systems that have proven efficiency in this context is the IDS. Scientists have long raced to develop different IDSs with tools including genetic algorithms (GA), k-NN, Bayesian, DT, fuzzy logic, and neural network classifiers to detect intrusions.¹¹ In this section, we will list some of these works.

In the work presented in [13], the authors implemented an IDS based on a back proliferation artificial neural network (ANN) to classify the network behaviour on KDD'99 dataset. The preparation and testing have been finished with 22 categories, and two styles have been thought of: training with and without normalization. Their models have achieved 85% and 79% detection rates, and 7% and 4% false alarm rates, with two normalization scenarios.¹² Manjula et al suggested a network attack detection model based on ML algorithms such as LR, Gaussian NB, SVM, and RF, on the NSLKDD dataset. Their trial results revealed that the RF classifier is superior compared to different techniques in detecting whether the information traffic is normal or an assault.¹³

A technique was introduced by M. A. Jabbar to upgrade the network assault recognition philosophy. This was done by consolidating the RF and the average one dependence estimator (AODE). The presented troupe classifier was dissected on Kyoto informational index, and inter quartile range (IQR) was utilized to eliminate clamor and anomalies in the information collection. For the analysis, 10-overlap cross-validation was employed to test the classifier, and 10 trees were utilized to build the RF classifier. The testing outputs were based on accuracy, detection rate, and false bug rate of 90.51%, 92.38%, and 0.14, respectively. It was noticed that the troupe classifier outperforms AODE, and Naïve Bayes classifiers.¹⁴

The authors of Reference 15 presented the component extraction estimation with the help of particle swarm optimization (PSO), GA optimization, gray wolf (GO) optimization, and firefly (FO) optimization. These techniques were applied on the UNSWNB15 dataset to find which included subset would create an ideal accuracy. After few investigations, a component subset containing 30 elements was chosen. The SVM and J48 tree models were utilized to execute the classification process. Their results illustrated that the J48 model achieved an accuracy of 90.484%, an FPR of 14.950%, and an F1_score of 90.172%. Furthermore, the SVM model achieved an accuracy of 90.119%, an FNR of 3.130%, an FPR 15.391%, and an F1_score of 89.808%.

In Reference 16, the authors proposed an IDS system that relies on a two-stage collection by bagging calculations and rotation forest technique. This multi-stage classifier used majority voting to select the best strategy. Furthermore, an element choice technique based on PSO, GA, and ant colony optimization (ACO) was introduced. The PSO-GA-ACO method was tested on the UNSW-NB15 dataset and 19 ideal features were extracated. To approve their two-stage technique, the authors involved the 10-overlap cross-validation method with the hold-out strategy. The technique proposed in this work achieved 91.27% and 91.60% accuracy and precision levels, respectively.

Khan et al¹⁷ introduced a feature selection strategy with the RF classifier to produce score and importance levels of every element in the UNSW-NB15 dataset. After a few trials, an element subset containing 11 categories was chosen. To perform the classification, the authors used the ML techniques including kNN, bagging meta estimator (BME), DT, RF, and XGBoost classifiers. Their model outcomes proved that the RF classifier gives the best outcomes with an accuracy of 75.56% and an F1_score of 73.00.

The authors of References 18,19 presented models built from scratch for IDSs based on stand-alone CNN architectures, and evaluated their models on the UNSW-NB15 dataset. The model structure in Reference 20 is composed of 5 convolutional layers giving 8, 16, 8, 16, and 32 feature maps, followed by 2 max-pool layers with a 2×2 pooling size, and 4 fully-connected (FC) layers. The authors achieved a binary classification accuracy of 89.26%. In Reference 21, the model structure was built with 3 convolutional layers, 3 max-pooling layers, 1 flatten, and 2 FC layers. The authors applied two techniques. The first is the extraction of the best features based on an information gain (IG) system, and the second is the application of a CNN to detect the novel types of attacks at the edge of the network. This method achieved a 91% classification accuracy.

The authors of References 22-24 adopted the idea of combining LSTM-RNN and CNN technology to build their IDSs with training from scratch and testing. To evaluate their models, they worked on the UNSW-NB15 IDS dataset. In Reference 22, both CNN and BiLSTM were employed to select the spatial and temporal features, respectively. The accuracy levels of classification achieved with the models introduced in References 22-24 were 77.16%, 87.70%, and 87.60%.

The volume of data has become very large in addition to the high price in the event of losing it. With the obvious shortcomings in the performance of traditional security systems, and the inefficiency and complexity of previous conventional ML methods and untrained (training from scratch) DL models for IDSs, we have developed automated IDSs using

CNNs and TL to overcome these problems. Our models showed a high ability to extract the best features and classify data automatically, effectively, and quickly in a simple way.

3 | PROPOSED MODELS BASED ON VGG19 AND RESNET152

In DL, a CNN is an ANN that is generally used for image classification. It is based on a shared-weight structure of the convolution filters through which all the entered attributes pass, leading to feature maps. CNNs are mainly composed of three kinds of layers, convolutional, pooling, and FC layers. Generally, a CNN extracts specific features using channels or filters, and in this way, pooling layers help the model in overlooking copied input. The produced features are passed to the FC layer for classification.

For some applications, the training information is less accessible. CNNs generally require a lot of training information to keep away from overfitting. A typical procedure is to train the network on a large dataset from a related space. When the network training is finished, an additional training stage is executed using the available limited data to calibrate or fine tune the weights, and this is called TL. Moreover, this method permits CNN models to effectively be applied on limited training sets.²⁰ TL is an undeniable level of ML, which has been comprehensively used in image portrayal. In this methodology, information on pre-trained models is moved to improve the learning in another task. The pre-trained TL models achieved high performance in image classification. TL is classified into three scenarios, shallow, deep, and fine-tuning. By fine-tuning, additional layers are trained, where the learning weights are tuned till a satisfactory performance is achieved.

As shown in Figure 2, the knowledge gained from the DL model built from scratch and trained on a large benchmark dataset is transferred to be used in the TL VGG19 and ResNet 152 models executed with parameter fine tuning. This process saves a lot of time that is lost in the training of the model from scratch. Our proposed TL models were tuned by modifying the hyper-parameters, such as example learning rate, activation function, number of filters, kernel size, loss function, optimizer type, and number of epochs. We continue these changes until a good performance is achieved.

3.1 | Proposed model based on VGG19

VGG is a CNN with a specific structure. It was introduced by a group of researchers at the University of Oxford. The VGG model was implemented using 14 million images, having a spot on the ImageNet dataset with 1000 classes. In our proposed VGG19 model, we deploy the VGG idea to be used in network IDS. The impact of deep convolutional neural networks (DCNNs) is transferred to the area of IDS. Small kernels of size 3×3 are utilized. The improvement in performance is gained from the expansion of the number of Conv layers from 13 in VGG16 to 16 in VGG19. The VGG19

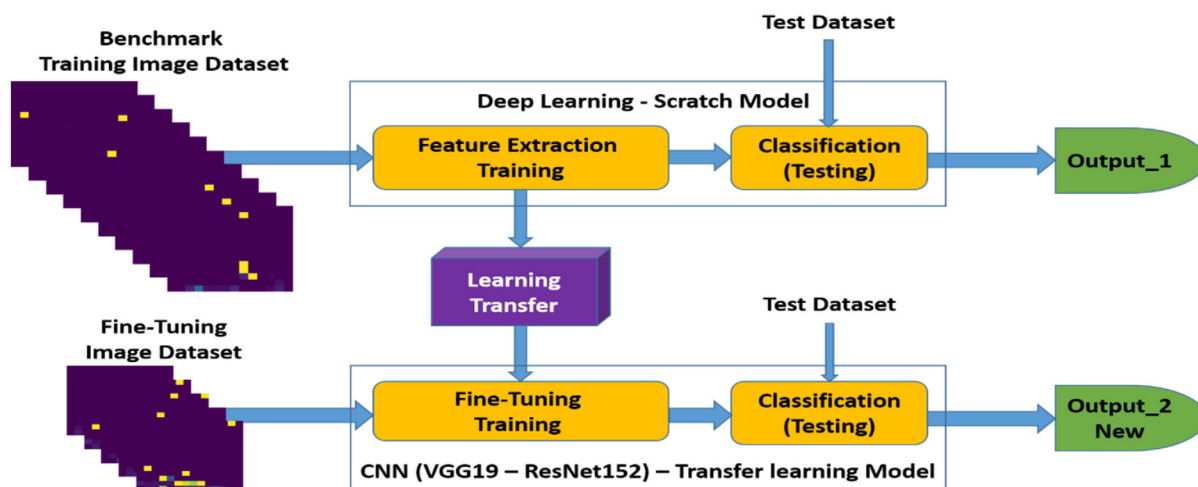


FIGURE 2 General block diagram of the TL process

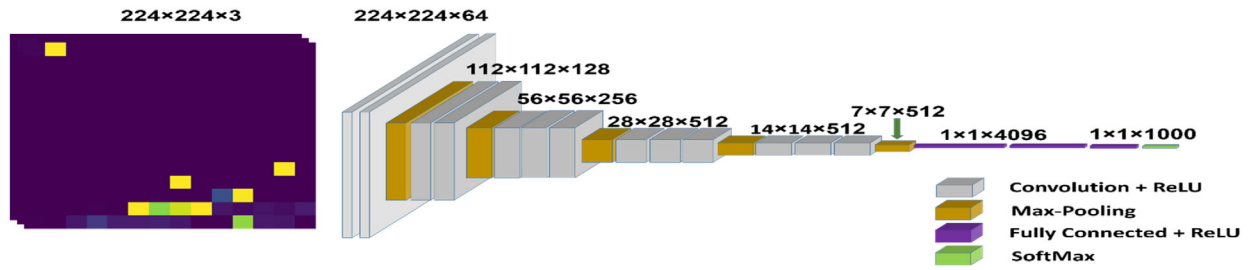


FIGURE 3 VGGNet-19 architecture



FIGURE 4 ResNet152 structure

architecture was built considering a 224×224 image size, 16 Conv layers (64, 128, 256, and 512 filters), 5 max-pool layers, 3 FC layer, Dropout 0.5, and 1 softmax layer.²¹ Figure 3 illustrates the architecture of the VGGNet-19 model.

3.2 | Proposed model based on ResNet152

Residual neural network (ResNet) has been presented in view of the possibility of residual learning for the simplicity of model training and lower complexity. Continuous learning capabilities concerned with layer inputs are utilized as opposed to learning of non-referential capabilities. Consequently, accuracy is enhanced as much as the model deepness is increased. ResNet was utilized for image portrayal in the challenge of 2015 ImageNet. It includes five stages, where each stage integrates one Conv block and one identity block. Each of these blocks consists of three Conv layers. After these five phases, subsampling is executed by an average pooling (Avg-Pool) layer. At the end, an FC layer employing a soft-max function is applied for data classification. ResNet152 is implemented with 152 layers.²⁵ Figure 4 shows the structure of the ResNet model. As shown in Figure 4, it was noticed that when the input of layer (i) is injected to the (i + j)th layer, the model training performance is enhanced, especially with the increase in the number of deep layers, and hence the model accuracy is also improved.

4 | RESULTS

4.1 | Dataset

UNSW-NB15 is an intrusion detection dataset. It contains nine unique assaults, incorporating DoS, analysis, backdoor, fuzzers, exploits, reconnaissance, generic, worms, and shellcode, along with normal behavior of data. It contains crude organization parcels. The training set has 175 341 records, while the testing set includes 82 332 records divided between clean data and attacks. The design of this dataset is more intricate in correlation with the other datasets like DARPA98, KDDCUP99, and NSL-KDD, which makes it more extensive for assessing the current IDSs in a more solid manner.²⁶

In our systems, we initially converted the dataset to be in the form of images. To do this task, we passed the original data through three stages. At first, we represented all types of data in a numerical representation. After that, one-hot-encoder is applied on this numerical data values to be represented between 0 and 1, which enabled us to overcome the problem

of large differences between the values of the data. In the last stage, each record is represented as an image. Figure 5A,B displays some samples of normal and abnormal network-practice record images.

4.2 | Model setup

Initially, the input images are resized to the size of 224×224 . Adam optimizer is utilized. Also, overfitting reduction is guaranteed. Softmax is applied at the final result layer. The number of epochs is 30 with VGG19, and 20 with ResNet152. After tuning of all hyper-parameters, the CNN models based on TL give a good classification performance on the UNSW-NB15 dataset.

The designs of the proposed models, are planned and implemented utilizing Kaggle, which offers free admittance to NVIDIA-TESLA-P100 GPUs notebook editors, along with 13 GB RAM working with Microsoft Windows 10 Pro 64bits. Python-3.7 is utilized for testing. In addition, Keras with tensor flow is also utilized for DL implementation.

4.3 | Evaluation metrics

In this work, we use several indicators to evaluate the efficiency of the proposed models, such as accuracy, recall or true positive rate (TPR), precision, F1_score, logarithmic loss, false alarm rate (FAR), in addition to the confusion matrix (CM). Moreover, ROC curve, AUC, and precision vs recall curve are considered. In addition, model complexity and execution time are considered. All TP, TN, and FP, and FN reflect the number of true and false predictions for the abnormal and normal behaviors of the data, respectively.²⁷ These metrics are calculated as per Equations (1) to (6).

1. Accuracy: It is given as the ratio of the number of true classifications to the total number of classifications.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \quad (1)$$

2. FAR: It represents the proportion of normal behaviors, which are wrongly classified as attacks.

$$\text{FAR} = \frac{\text{FP}}{\text{FP} + \text{TN}}. \quad (2)$$

3. TPR, recall, or sensitivity: It refers to the probability of a positive action, classified correctly as being positive.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (3)$$

4. Precision: It expresses the data that the system has indicated that it is associated with a specific category, and it is already associated with it.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \quad (4)$$

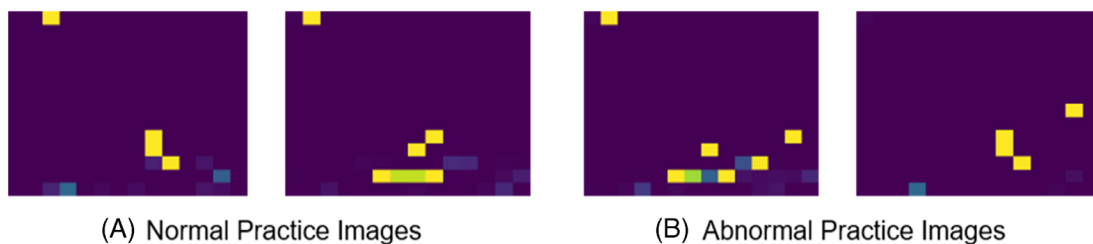


FIGURE 5 Normal and abnormal network-practice image samples

5. F1-score: It is the metric, which adds recall to precision in one single value. It allows to compare properly between different models with different precision and recall values.

$$F1_score = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}. \quad (5)$$

6. Logarithmic loss: It reflects the closeness between the prediction probability and its real value.

$$\log \text{ loss} = -\log P \left(\frac{yT}{yP} \right) = -yT \log(yP) + (1 - yT) \log(1 - yP). \quad (6)$$

7. Confusion matrix:

Anomaly	TP	FN
Normaly	FP	TN
	Anomaly	Normaly

8. ROC curve: It refers to the graph of TPR vs FPR, which highlights the model performance at all thresholds of classification.
9. Recall vs precision curve: This curve indicates the tradeoff between recall and precision at different thresholds.
10. Complexity and execution time: They represent the computational cost required by the models.

4.4 | Obtained results

Based on the UNSW-NB15 dataset, 10 000 images representing normal and abnormal network behavior were used to train and test our models. This dataset was divided into different percentages for train and testing, as 60/40, 70/30, and 80/20.

In terms of the classification accuracy and log loss, it was observed that the highest values of 94.05% and 2.0524, respectively, were achieved with our ResNet152 model. These results were obtained with 80/20 train to test ratio. According to the precision, recall, and F1_score, the classification models were evaluated based on three scenarios:

1. If the increase of FP errors is not accepted, the model with high precision value is selected.
2. If the increase of FN errors is not accepted, the model with high recall value is selected.
3. If both FN and FP errors are not accepted, the selection relies on the highest F1_score value.

Our ResNet152 has achieved the same value of 0.99 for precision, recall, and F1_score, which is very close to 1. This makes the model suitable for all of the above-mentioned evaluation scenarios. These values prove that the model predicts real events correctly, which means that the model predictions relate the data efficiently to its correct category. Comparing the two proposed models, the IDS-ResNet152 model achieves equal values of recall, precision, and F1_score, which makes it better for use compared to the IDS-VGG19 model.

According to the ROC curve, the AUC value always lies between 0 and 1. In our IDSs based on ResNet152 and VGG19 models, the AUC values are 0.92 and 0.90, respectively. The summary of numerical results is shown in Table 1. In addition, the recall vs precision curves are given in Table 1. The obtained results reveal the effectiveness and reliability of the proposed models according to their low positive and negative error rates at the same time.

Our proposed models were built on the theory of TL, which relies on fine-tuning training of the model on a small dataset without the need to train the model from scratch. This technique enables to get the knowledge gained from previously-trained DL models and use it in our models. As a result, our models are simple and they save the very long time required to train the model from scratch (Tables 1 and 2).

4.5 | Comparison of results

In Table 3, we compared our proposed IDSs based on TL models with some of the previous models, which were built based on the conventional ML theories and untrained DL-CNN models. Experiments have been implemented on UNSW-NB15

TABLE 1 Illustrative binary-classification results of the ResNet152 and VGG19 models

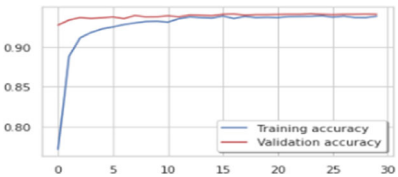
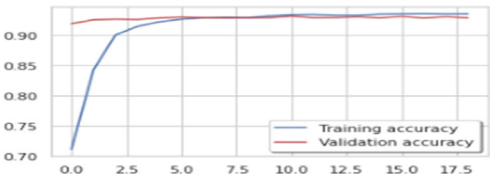
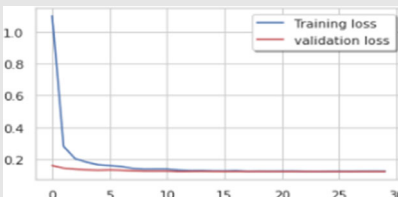
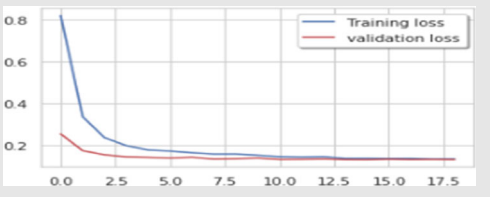
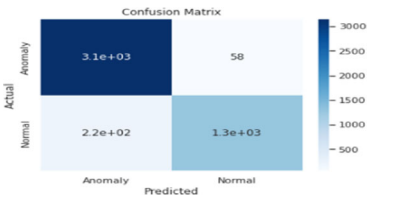
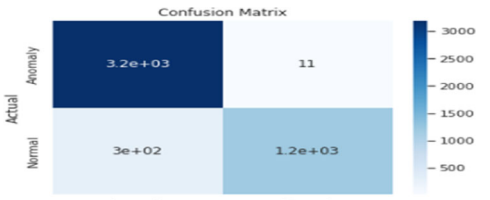
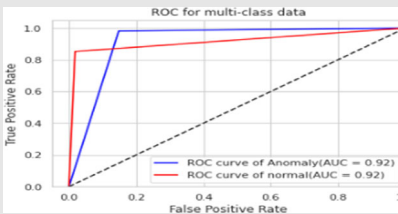
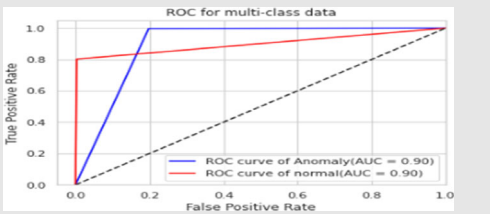
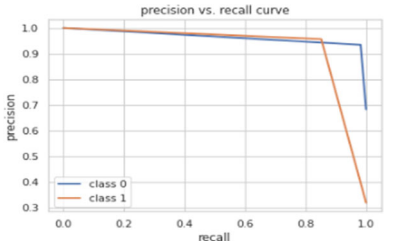
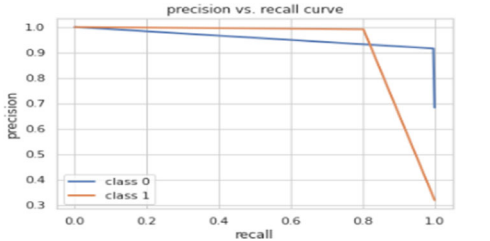
Model	Binary classification	
	VGG19	ResNet152
Accuracy No. of epochs on x-axis Accuracy on y-axis		
Loss No. of epochs on x-axis loss on y-axis		
CM		
ROC		
Recall vs precision		

TABLE 2 Numerical binary-classification results of the VGG19 and ResNet152 models

Model	Image dimensions	Train/Test	Accuracy (%)	Loss	Precision (%)	Recall (%)	F1_score (%)	Log loss
ResNet152	(224, 224, 3)	80/20	94.05	0.1230	95	92	93	2.0524
		70/30	93.90	0.1218	95	91	93	2.1041
		60/40	93.94	0.1225	99	99	99	2.0929
VGG19	(224, 224, 3)	80/20	93.48	0.1248	95	90	92	2.2510
		70/30	93.48	0.1260	95	90	92	2.2512
		60/40	93.56	0.1247	95	90	92	2.2216

TABLE 3 Comparison between our proposed models and other related IDS models

Authors	Year	Dataset	Technique	Binary classification			Feature extraction and classification in one process (Yes or No)			Training type	Computational complexity/ Processing time (seconds)	
				accuracy %	Precision %	Recall %	F1_score %	Train	Test			
Almomani ¹⁵	2020	UNSW-NB15	J48	90.48	84.136	97.141	90.172	No	Training from scratch	-	-	
			SVM	90.11	83.706	96.870	89.808					
Tama et al ¹⁶	2019	UNSW-NB15	PSO-GA	91.27	91.60	91.30	91.45	No	Training from scratch	-	-	
Khan et al ¹⁷	2018	UNSW-NB15	RF	75.56	-	-	-	No	Training from scratch	-	-	
Al-Turaiki et al ¹⁸	2021	UNSW-NB15	CNN-DFS	89.26	89	89	89	No	Training from scratch	872.4	229.2	
Gamal et al ¹⁹	2020	UNSW-NB15	CNN	91.00	-	-	-	No	Training from scratch	-	-	
Jiang et al ²²	2020	UNSW-NB15	CNN + LSTM	76.82	81.48	78.47	79.95	No	Training from scratch	2750.47	-	
Cai et al ²³	2022	UNSW-NB15	CNN + LSTM	87.70	88.46	87.70	85.44	No	Training from scratch	-	-	
Meliboev et al ²⁴	2022	UNSW-NB15	CNN + LSTM	87.60	85.5	90.6	88	No	Training from scratch	-	-	
Youssef et al	Proposed model_1	UNSW-NB15	ResNet152 (CNN)	93.94	99	99	99	Yes (automated)	Fine-tuning	1220	5.5	
Youssef et al	Proposed model_2	UNSW-NB15	VGG19 (CNN)	93.56	95	90	92	Yes (automated)	Fine-tuning	1830	7.2	

dataset, considering classification accuracy, type of technique, the process of feature extraction, and the model training methodology. From the comparison, it was found that our models outperform the others in terms of classification accuracy. Tama et al¹⁶ recorded the highest values compared to all previous stated models with an accuracy value of 91.27%, but our models achieved accuracy levels of 93.94% and 93.56%. On the other hand, the previous models were trained from scratch, where a lot of time and resources were wasted. The traditional ML techniques add one more separate phase for manual feature extraction, which increases the model complexity and cost. Our models gain the benefit of automated feature extraction by CNNs, and the benefit of TL in transferring the knowledge without any need to spend more time in training from scratch. These benefits make our models simpler with less computations, and make them more suitable to be applied in online intrusion detection due to their low required execution times. Even though the model in Reference 18 has a training time of 872.4, which is less than those of our proposals, one of our models achieved a smaller test time of 5.5 s compared to the 229.2 s recorded in Reference 18. These achievements make our models better for online anomaly detection.

5 | CONCLUSION AND FUTURE WORK

Proceeding from the great importance of data and communication networks, there is a large development in security methods, in order to confront the continuous advancement in cyber-attacks philosophy. In this work, we have developed automated anomaly-based techniques for detecting network intrusions, by applying the VGG19 and ResNet152 models with TL. To evaluate the proposed models, the UNSW-NB15 modern IDS dataset has been used. The performance assessment is implemented based on the classification accuracy, recall, precision, F1_score, loss, log loss, FAR, TPR, ROC and AUC, and confusion matrix. In addition, we compared our models with some related newly-designed ones based on the classification accuracy, technique used, complexity, training method, and the processing time. The results revealed the superiority of the proposed models. In the future work, we will test our proposed models on other modern network datasets, and develop other models for new types of attacks.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interests.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

This article does not contain any studies with human participants or animals performed by any of the authors.

ORCID

Youssef F. Sallam  <https://orcid.org/0000-0003-1817-3543>

Walid El-Shafai  <https://orcid.org/0000-0001-7509-2120>

Nirmeen A. El-Bahnasawy  <https://orcid.org/0000-0002-4542-323X>

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